

Decentralised Adapter-Based LLM Inference: A Systematic Literature Review

Mapping the research gap at the intersection of PEFT • P2P Systems • Multi-Task Serving

PROBLEM

Full fine-tuning is prohibitively expensive

Serving systems assume centralised orchestration

Can a P2P direction work?



CENTRAL QUESTION

Can task-specific adapters be...

discovered

retrieved

composed

served

...across autonomous nodes sharing a **frozen backbone**, without a central coordinator?



METHOD

PRISMA 2020 + Wohlin Snowballing

1,150 candidates → 972 screened
merged with 352 pre-validated corpus

123 included records (120 distinct papers)
5 clusters • 6-dimension rubric • Gap map



KEY FINDING

No system simultaneously addresses:

Frozen shared backbone

Adapter-level exchange

P2P topology

Decentralised discovery

Multi-task fusion

No central coordinator

→ Partitions into four gap quadrants

THE FOUR GAP QUADRANTS

CONCEPTUAL

What P2P adapter architectures are possible?

ALGORITHMIC

How to discover, route, and compose adapters?

SYSTEMS

How to build decentralised inference systems?

EMPIRICAL

How does it perform in practice?

Establishes the evidence base and research agenda for decentralised, adapter-based P2P LLM inference

